

A Sharp Ellipse Dichotomy for a Complex Busemann–Petty Problem

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Status. Candidate full counterexample theorem, likely valid, pending expert review. For every $-2 < p < 0$, the Busemann–Petty implication for the complex L_p -intersection operator in $\mathbb{C}^2$ is affirmative for circular planar ellipses and negative for every noncircular origin-centered ellipse.

1 Source problem and the untreated case

Ellmeyer and Hofstätter [1] define complex L_p -intersection operators $I_{C,p}$ for an origin-symmetric planar convex body $C \subset \mathbb{C}$ and ask their general Busemann–Petty Problem 1: for K, L in the injectivity set $\mathcal{K}(I_{C,p})$, does

$$I_{C,p}K \subseteq I_{C,p}L \implies V_4(K) \leq V_4(L) \quad (1)$$

hold? The source proves that for $-2 < p < 0$ the answer in complex dimension two is affirmative when C is the disk. Its main theorem does not treat general C in that dimension; the introduction explains that the image is described completely only for the disk.

The result below fills this gap for every origin-centered ellipse. It also shows that the disk theorem is unstable: an arbitrarily small noncircular elliptic deformation reverses the answer.

2 Source identities and notation

Write $m = 4$ and let $\mathbf{F}_q$ be the spherical Fourier transform of degree q on even functions on $\mathbb{S}^3$, with the normalization of [1]. For a finite measure ν on $\mathbb{S}^1$, set

$$(T_\nu f)(u) = \int_{\mathbb{S}^1} f(cu) d\nu(c).$$

The source proves that

$$\rho_{I_{C,p}K}^{-p} = \frac{1}{(2\pi)^2(4+p)} T_{\nu_{C,p}} \mathbf{F}_{-4-p} \rho_K^{4+p}, \quad \nu_{C,p} = \mathbf{F}_p h_C^p, \quad (2)$$

where $\nu_{C,p}$ is a positive nonzero even measure for $p < 0$. It also proves

$$\mathbf{F}_{-4-p} \mathbf{F}_p f = (2\pi)^4 f, \quad (3)$$

and that the transforms are self-adjoint.

$$\rho_{IK}(u) = V_{n-1}(K \cap u^\perp), \quad u \in \mathbb{S}^{n-1}.$$

The projection body ΠK of a convex body $K \in \mathcal{K}(\mathbb{R}^n)$ is defined by (see, e.g., [9])

$$h_{\Pi K}(u) = V_{n-1}(K|u^\perp), \quad u \in \mathbb{S}^{n-1},$$

where $h_K(u) = \sup\{\langle x, u \rangle : x \in K\}$, $u \in \mathbb{S}^{n-1}$, is the support function of K , and $\langle \cdot, \cdot \rangle$ is the Euclidean inner product.

In this way, the Busemann–Petty problem and Shephard’s problem are special cases of the following more general class of Busemann–Petty type problems which we vaguely formulate for geometric operators on (subsets of) convex bodies with non-empty interiors and make more precise later.

Problem 1. *Let Φ be a geometric operator and let $K, L \in \mathcal{K}(\Phi)$.*

$$\text{Does} \quad \Phi K \subseteq \Phi L \quad \implies \quad V_n(K) \leq V_n(L) \quad \text{hold?}$$

Here, we denote by $\mathcal{K}(\Phi) \subset \mathcal{K}(\mathbb{R}^n)$ a subset of convex bodies with non-empty interior, for which we will consider Problem 1. Usually, $\mathcal{K}(\Phi)$ will be the injectivity set of the operator Φ , which turns out to be the largest subset for which Problem 1 is non-trivial. Problem 1 has been considered widely, e.g., for centroid bodies [28], for L_p -intersection bodies [44], for Minkowski valuations compatible with rotations [40], for complex intersection bodies [22], replacing real by complex hyperplanes in [11] and origin-symmetry by $\mathbb{S}^1$ -invariance (that is, invariance under multiplication by $c \in \mathbb{S}^1 \subset \mathbb{C}$), as well as in a functional setting [23].

In this work, we consider Problem 1 for the recently defined analogues of projection

Figure 1: Problem 1 on page 2 of arXiv:2404.05630v3.

The injectivity set is characterized by the angular Fourier coefficients: an origin-symmetric body belongs to $\mathcal{K}(\mathcal{I}_{C,p})$ exactly when every harmonic component of ρ_K^{4+p} corresponding to a zero coefficient of h_C^p vanishes. In particular, if all even Fourier coefficients of h_C^p are nonzero, $\mathcal{K}(\mathcal{I}_{C,p})$ is the set of all origin-symmetric convex bodies with nonempty interior.

Finally, for star bodies in $\mathbb{R}^4$ the dual mixed volume is

$$\tilde{V}_{-p}(K, L) = \frac{1}{4} \int_{\mathbb{S}^3} \rho_K^{4+p} \rho_L^{-p},$$

and the dual L_p -Minkowski inequality used in [1] says

$$-p\tilde{V}_{-p}(K, L) \leq -pV_4(K)^{(4+p)/4}V_4(L)^{-p/4}. \quad (4)$$

3 Proof intuition

The disk kills every non- $\mathbb{S}^1$ -invariant angular harmonic, so its injectivity set contains only balanced bodies. A noncircular ellipse does the opposite: every even angular multiplier is nonzero, hence all origin-symmetric bodies are admissible.

This permits a sign-changing perturbation. On a real Lagrangian plane $E \subset \mathbb{C}^2$, choose an even function φ which is negative on $\mathbb{S}^3 \cap E$, but whose positive angular convolution $T_{\nu_{C,p}}\varphi$ is strictly positive everywhere. A thin real ellipsoid has L_p -embedding measure concentrated on that circle, so the pairing of φ with the measure is negative. Perturbing the radial power of the ellipsoid by $\mathbf{F}_p\varphi$ makes the transformed body smaller pointwise, while the dual mixed-volume calculation makes its ordinary volume strictly larger. These are exactly the two inequalities needed for a counterexample.

polar of the complex centroid body from [14]. Moreover, the real operators can be recovered by letting $C = [-1, 1] \subset \mathbb{C}$ and $p \rightarrow -1^+$ (up to suitable renormalizations). In the following, we consider the operators Π_C and $I_{C,p}$ only for origin-symmetric $C \in \mathcal{K}(\mathbb{C})$ containing the origin in its interior. This will be a general condition in all statements below, together with the assumption that $-2 < p \leq 1$ is non-zero. As in the real setting, Π_C and $I_{C,p}$ behave in a dual way and the roles of the bodies K and L in Problem 1 are interchanged. In order to state the results in a unified way, we will therefore "dualize" the problem for Π_C by considering its polar operator Π_C° , $\Pi_C^\circ K = (\Pi_C K)^\circ$, where $K^\circ = \{y \in \mathbb{R}^m : \langle y, x \rangle \leq 1, \forall x \in K\}$ denotes the polar body of $K \in \mathcal{K}(\mathbb{R}^m)$ containing the origin in its interior. As polarity is order-reversing, this interchanges K and L in Problem 1 but has no further effects. Moreover, note that for $p > 0$, the volume inequality that we expect in Problem 1 is reversed. We will take care of this fact by multiplying both sides with $-p < 0$ and set $p = 1$ when $\Phi = \Pi_C^\circ$. Let $\mathbb{D} \subset \mathbb{C}$ denote the unit disc. Our main result states the following.

Theorem A. *Let $\Phi \in \{\Pi_C^\circ, I_{C,p}\}$. If $p < 0$, then the answer to Problem 1 is affirmative for $n = 2$ and $\Phi = I_{\mathbb{D},p}$, and negative for $n \geq 3$. If $p > 0$, then the answer to Problem 1 with reversed volume inequality is negative for all $n \geq 2$.*

Let us point out that Theorem A implies a similar statement for the complex centroid body defined in [14] (see Remark 3.6 below).

Our proof of Theorem A follows a strategy similar to [10, 24, 37, 44] for real Busemann–Petty type problems. In particular, it relies heavily on a connection to the spherical Fourier transform. Indeed, all operators $\Phi \in \{\Pi_C^\circ, I_{C,p}\}$ can be represented in terms

Figure 2: Theorem A on page 3 singles out the disk in complex dimension two.

4 The ellipse dichotomy

Theorem 1 (sharp dichotomy among ellipses). *Fix $-2 < p < 0$, and let $C \subset \mathbb{C}$ be an origin-centered ellipse with nonempty interior.*

- (i) *If C is a disk up to dilation, implication (1) holds.*
- (ii) *If C is noncircular, there are smooth origin-symmetric strictly convex bodies $K, L \subset \mathbb{C}^2$ in $\mathcal{K}(I_{C,p})$ such that*

$$I_{C,p}K \subsetneq I_{C,p}L \quad \text{and} \quad V_4(K) > V_4(L).$$

Thus the disk is the unique affirmative ellipse for every exponent in the entire interval $(-2, 0)$.

Part (i) is Theorem A and Proposition 1.5 of [1]; dilating C merely rescales the operator. We prove part (ii) in four steps.

4.1 Every even ellipse multiplier is nonzero

After a rotation and dilation, the support function of a noncircular ellipse satisfies

$$h_C(e^{i\theta})^2 = A + B \cos(2\theta), \quad A > |B| > 0, \quad B \neq 0.$$

Put $s = -p/2 > 0$. For $j \geq 0$, the Fourier coefficient at frequency $2j$ of h_C^p is, up to a nonzero normalization, the coefficient at frequency j of $(A + B \cos x)^{-s}$. The Laplace representation gives

$$(A + B \cos x)^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-At} e^{-Bt \cos x} dt,$$

$$\widehat{h_C^p}(2j) = \frac{(-\operatorname{sgn} B)^j}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-At} I_j(|B|t) dt,$$

again up to a fixed positive normalization. The modified Bessel function $I_j(t)$ is strictly positive for $t > 0$, so this coefficient never vanishes. Origin symmetry kills only the odd coefficients, which are irrelevant for even data. Consequently $\mathcal{I}_{C,p}$ is injective on every even function and

$$\mathcal{K}(\mathcal{I}_{C,p}) = \{\text{all origin-symmetric convex bodies with nonempty interior}\}. \quad (5)$$

4.2 A sign-changing function with positive angular image

The same ellipse can be written $h_C(x) = |Bx|$ on $\mathbb{R}^2$ for an invertible real matrix B . The standard Fourier formula for homogeneous Euclidean powers [2, Chapter 3] and a linear change of variables give

$$d\nu_{C,p}(c) = \kappa_p |\det B|^{-1} |B^{-T}c|^{-2-p} dc, \quad \kappa_p > 0. \quad (6)$$

Thus $\nu = \nu_{C,p}$ has a positive continuous density on $\mathbb{S}^1$. Let

$$M = \nu(\mathbb{S}^1), \quad r = \frac{1}{M} \left| \int_{\mathbb{S}^1} c^2 d\nu(c) \right| < 1.$$

The strict inequality follows from equality in the triangle inequality: since the density is positive everywhere, c^2 is not constant on its support.

Take the real Lagrangian plane $E = \mathbb{R}^2 \subset \mathbb{C}^2$ and define

$$Q(u) = |P_E u|^2, \quad u \in \mathbb{S}^3.$$

Writing $u = x + iy$ with $x, y \in E$ and $c = e^{i\theta}$,

$$Q(cu) = \frac{1}{2} + \frac{|x|^2 - |y|^2}{2} \cos(2\theta) - \langle x, y \rangle \sin(2\theta).$$

The amplitude of the second harmonic is at most $1/2$, because

$$(|x|^2 - |y|^2)^2 + 4\langle x, y \rangle^2 \leq (|x|^2 + |y|^2)^2 = 1.$$

It follows that

$$T_\nu Q(u) \leq \frac{M(1+r)}{2} \quad (u \in \mathbb{S}^3). \quad (7)$$

Choose

$$1 < a < \frac{2}{1+r}, \quad \varphi = 1 - aQ. \quad (8)$$

Then φ is smooth and even,

$$T_\nu \varphi \geq M \left(1 - \frac{a(1+r)}{2} \right) > 0, \quad \varphi|_{\mathbb{S}^3 \cap E} = 1 - a < 0. \quad (9)$$

4.3 An embedding measure concentrated on the negative circle

Let $F = iE$ and, for $0 < \delta < 1$, set

$$A_\delta = \delta P_E + P_F, \quad L_\delta = A_\delta B_2^4.$$

This is a smooth strictly convex origin-symmetric ellipsoid. Since $\rho_{L_\delta}^{-p}(u) = \|u\|_{L_\delta}^p$, another linear change in the Fourier transform yields the positive density

$$d\mu_\delta(u) := d(\mathbf{F}_p \rho_{L_\delta}^{-p})(u) = \gamma_p \delta^2 (\delta^2 |P_E u|^2 + |P_F u|^2)^{-(4+p)/2} du, \quad \gamma_p > 0. \quad (10)$$

The normalized measures $\mu_\delta/\mu_\delta(\mathbb{S}^3)$ concentrate on $\mathbb{S}^3 \cap E$ as $\delta \downarrow 0$. Here is a direct check. Under uniform measure on $\mathbb{S}^3$, the variable $t = |P_E u|^2$ is uniform on $[0, 1]$; hence, up to a positive constant, the density in t is

$$\delta^2(\delta^2 t + 1 - t)^{-\alpha}, \quad \alpha = \frac{4+p}{2} \in (1, 2).$$

Outside any fixed neighborhood of $t = 1$ its mass is $O(\delta^2)$. On $1 - \delta^2 \leq t \leq 1$ its mass is bounded below by a constant times $\delta^{4-2\alpha} = \delta^{-p}$. Since $2 + p > 0$, the ratio tends to zero. This proves concentration. By (9), therefore,

$$\frac{\langle \varphi, \mu_\delta \rangle}{\mu_\delta(\mathbb{S}^3)} \longrightarrow 1 - a < 0. \quad (11)$$

Fix δ small enough that $\langle \varphi, \mu_\delta \rangle < 0$, and write $L = L_\delta$.

4.4 The radial perturbation

Let $\psi = \mathbf{F}_p \varphi$. For $\varepsilon > 0$ define a star body K by

$$\rho_K^{4+p} = \rho_L^{4+p} - \varepsilon \psi. \quad (12)$$

For sufficiently small ε , the right side is positive and K is smooth and strictly convex. This is the standard radial perturbation lemma used as Lemma 3.1 in [1]; strict positive curvature is open under the resulting C^2 perturbation of the ellipsoid. Both bodies are origin-symmetric, so (5) puts K, L in the required injectivity set.

Apply (2), (3), and (9):

$$\rho_{I_{C,p}K}^{-p} = \rho_{I_{C,p}L}^{-p} - \varepsilon \frac{(2\pi)^2}{4+p} T_\nu \varphi < \rho_{I_{C,p}L}^{-p}.$$

Because $-p > 0$, this is equivalent to the strict radial inclusion $I_{C,p}K \subsetneq I_{C,p}L$.

For the volume sign, self-adjointness of $\mathbf{F}_p$ and (11) give

$$\begin{aligned} -4p\tilde{V}_{-p}(K, L) &= -p\langle \rho_K^{4+p}, \rho_L^{-p} \rangle \\ &= -4pV_4(L) + p\varepsilon \langle \psi, \rho_L^{-p} \rangle \\ &= -4pV_4(L) + p\varepsilon \langle \varphi, \mathbf{F}_p \rho_L^{-p} \rangle > -4pV_4(L). \end{aligned}$$

Combining this strict inequality with (4) gives

$$-pV_4(L) < -pV_4(K)^{(4+p)/4} V_4(L)^{-p/4},$$

and hence $V_4(K) > V_4(L)$. This proves Theorem 1.

5 Verification, scope, and novelty

The proof is analytic; no numerical assertion is used. The included script `code/numerical_sanity.py` evaluates the explicit one-dimensional second-moment and concentration integrals for one ellipse and five exponents. In each case it confirms both strict signs in (9) and (11). It is only a regression check.

The claim fully classifies origin-centered ellipses, but does not classify all origin-symmetric planar bodies C . The same argument works whenever all even coefficients of h_C^p are nonzero and the

second angular moment is strict. Kernels with finite rotational symmetry can have systematic zero multipliers; then the concentrating real ellipsoid need not belong to the injectivity set, so a new symmetry-compatible construction is needed.

The novelty search was bounded but current through August 11, 2026. It covered the run indexes; exact arXiv-id, title, author, ellipse, and noncircular-ellipse searches; the current arXiv v3 and published-paper metadata; the authors' publication list; and their MathOverflow errata page. No later resolution or ellipse dichotomy was found. The mechanism uses the source's Fourier and perturbation framework in a new non- $\mathbb{S}^1$ -invariant way, so expert novelty review remains appropriate.

Human review should prioritize four points: the nonvanishing Bessel coefficient calculation; the normalization-independent strict bound (7); the weak concentration in (10); and the two signs in the final dual mixed-volume calculation.

References

- [1] S. Ellmeyer and G. C. Hofstätter, *Busemann–Petty type problems on complex vector spaces*, Indiana Univ. Math. J. **75** (2026), no. 2, 525–551; arXiv:2404.05630v3.
- [2] A. Koldobsky, *Fourier Analysis in Convex Geometry*, Mathematical Surveys and Monographs 116, American Mathematical Society, 2005.